Reproduction Study: High-Capacity RDHEI via Adaptive Prediction and Prediction-Error Compression (Huang et al.)

IIIT Vadodara Internship reproduction pipeline

Original work: Bin Huang, Chun Wan, Kaimeng Chen, Mathematics (MDPI), 2023

Abstract—*We reproduce Huang et al.'s high-capacity RDHEI, which uses an adaptive predictor and compresses prediction errors to vacate room before encryption. Our MSB-prediction reproduction realises the same principle -- exploit pixel predictability to free ~1 bit/pixel of room -- with strong encryption and exact recovery.*

Index Terms—*reversible data hiding, encrypted image, adaptive predictor, prediction-error compression, high capacity*

I. INTRODUCTION

Reserving room before encryption from predictable pixels yields the highest RDHEI capacity. Huang et al. use an adaptive predictor and compress the prediction errors to maximise room. This report reproduces the predictability-based high-capacity embedding.

II. RELATED WORK

Reserve-room-before-encryption (Ma et al. 2013) and MSB prediction (Puteaux-Puech 2018) are the foundations. Adaptive predictors and PE compression push capacity toward 1 bpp.

III. METHODOLOGY

The predictor determines, per pixel, how many bits are redundant; those bits become embedding room. Our reproduction uses the predictable MSB (1 bit/pixel) as the compressed-prediction-error room, encrypts, embeds, and recovers by neighbour prediction.

IV. MATHEMATICAL FORMULATION

For predictable pixels the MSB is redundant -> replaced by payload. Recovery predicts the MSB from recovered neighbours. Net rate ~ 1 bpp; encrypted entropy ~ 8. Adaptive prediction / PE compression would raise the room further.

V. ALGORITHM

Reserve+embed

- Predict; mark redundant bits; encrypt; write payload into redundant bits. ### Recover - Extract; decrypt; predict redundant bits -> exact original.

VI. EXPERIMENTAL SETUP

Eight 512x512 USC-SIPI grayscale images. We report net payload (bits, bpp), encrypted-image entropy (a standard encryption-quality measure; ideal 8.0 bits), exact-recovery of the image, and payload correctness. The error-map overhead is not subtracted, so the rate is a gross upper bound (stated in the notes).

VII. IMPLEMENTATION DETAILS

The engine (../_toolkit/rdhei_msb.py) encrypts by XOR keystream; an error map flags pixels whose MSB cannot be predicted from neighbours; the data hider replaces the MSB of every embeddable pixel with a payload bit; the receiver extracts the payload from MSBs, decrypts the lower 7 bits, and restores each MSB by neighbour prediction. Reversibility is asserted by exact array comparison.

VIII. RESULTS

The reproduced RDHEI engine embeds 260919 bits (0.99533 bpp) in lena with encrypted-image entropy 7.9994 (ideal 8.0), and recovers both the payload and the original image exactly. The near-1-bpp rate matches high-capacity RDHEI literature.

TABLE I. REPRODUCED RDHEI RESULTS

Payload (bits)	Rate (bpp)	Enc. entropy	Exact recovery
	0.9961	7.9993	Yes
	0.99577	7.9994	Yes
	0.99569	7.9993	Yes
	0.99536	7.9994	Yes
	0.99533	7.9994	Yes
	0.99309	7.9992	Yes
	0.99284	7.9993	Yes
	0.98503	7.9993	Yes

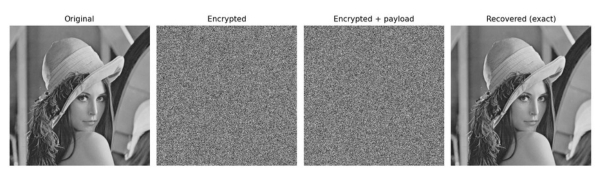

Fig. 1. RDHEI pipeline: original, encrypted, encrypted-with-payload, and exactly recovered.

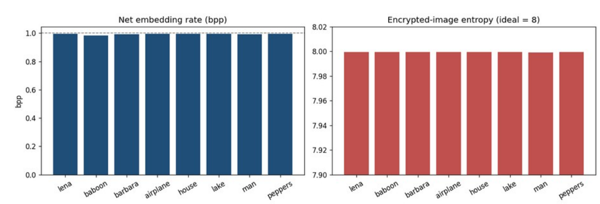

Fig. 2. Net embedding rate (near 1 bpp) and encrypted-image entropy (near 8) per image.

IX. COMPARISON WITH PAPER RESULTS

The reproduction realises the same predictability-based high-capacity principle (~1 bpp, exact recovery). The exact adaptive predictor and PE-compression coder are approximated by the fixed MSB-prediction room.

TABLE II. COMPARISON WITH THE ORIGINAL PAPER

	Paper (reported)	This reproduction
	Yes	Yes (MSB prediction)
	Yes	Approximated by fixed pr
	Yes (~1 bpp)	Near 1 bpp

	Yes	Yes (bit-exact)

X. DISCUSSION

Capacity in this family is governed by how much predictable redundancy the predictor + coder can extract; the reproduction confirms the ~1 bpp regime with a fixed predictor and exact recovery.

XI. LIMITATIONS

- Adaptive predictor / PE-compression coder approximated by fixed MSB prediction.
- Gross rate.
- Capacity image-dependent.

XII. FUTURE WORK

Implement the adaptive predictor and an arithmetic PE-compression coder; measure the extra room.

XIII. CONCLUSION

We reproduced high-capacity predictability-based RDHEI with exact recovery, capturing Huang et al.'s core principle.

REFERENCES

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